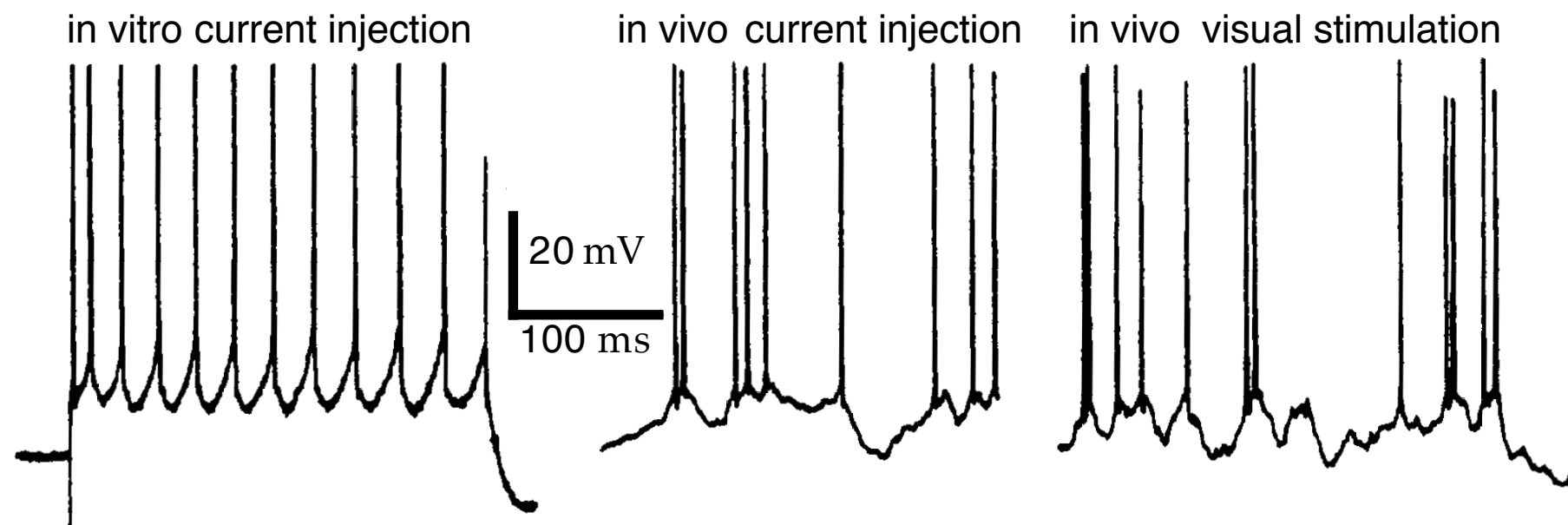
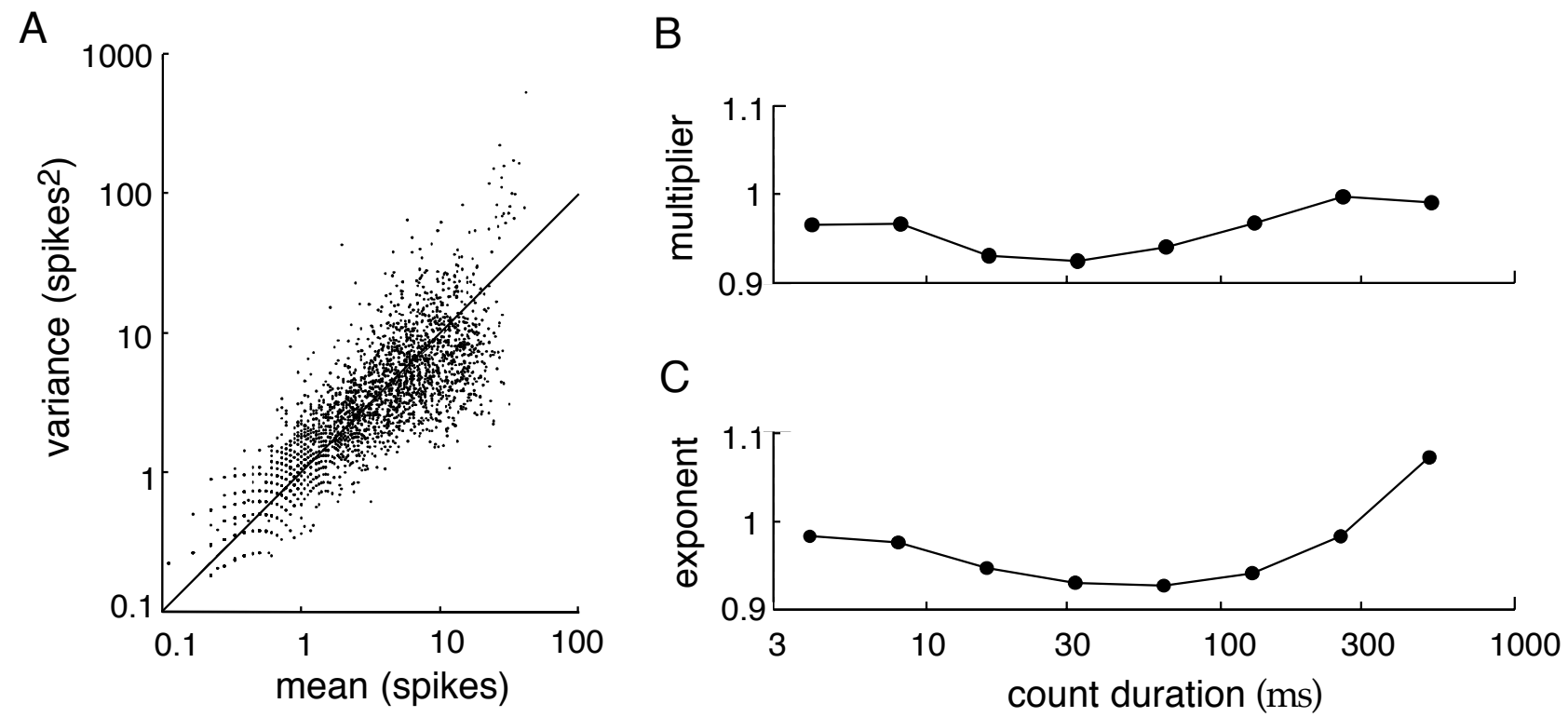


E-I balanced neural dynamics

The Poisson spike train



The Poisson spike train



The Poisson spike train

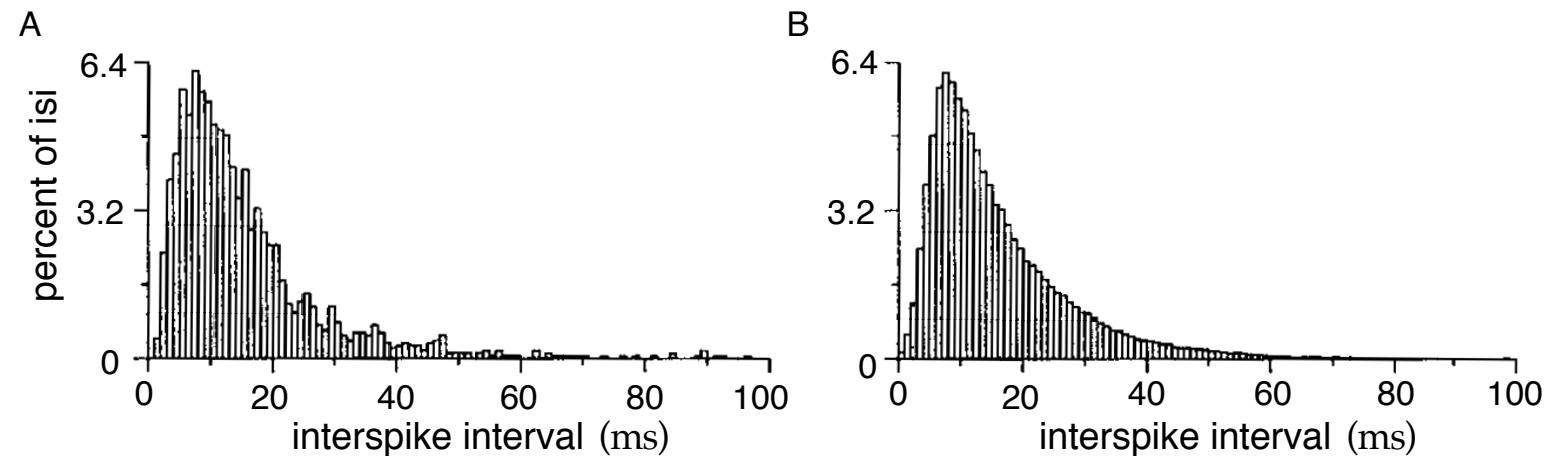
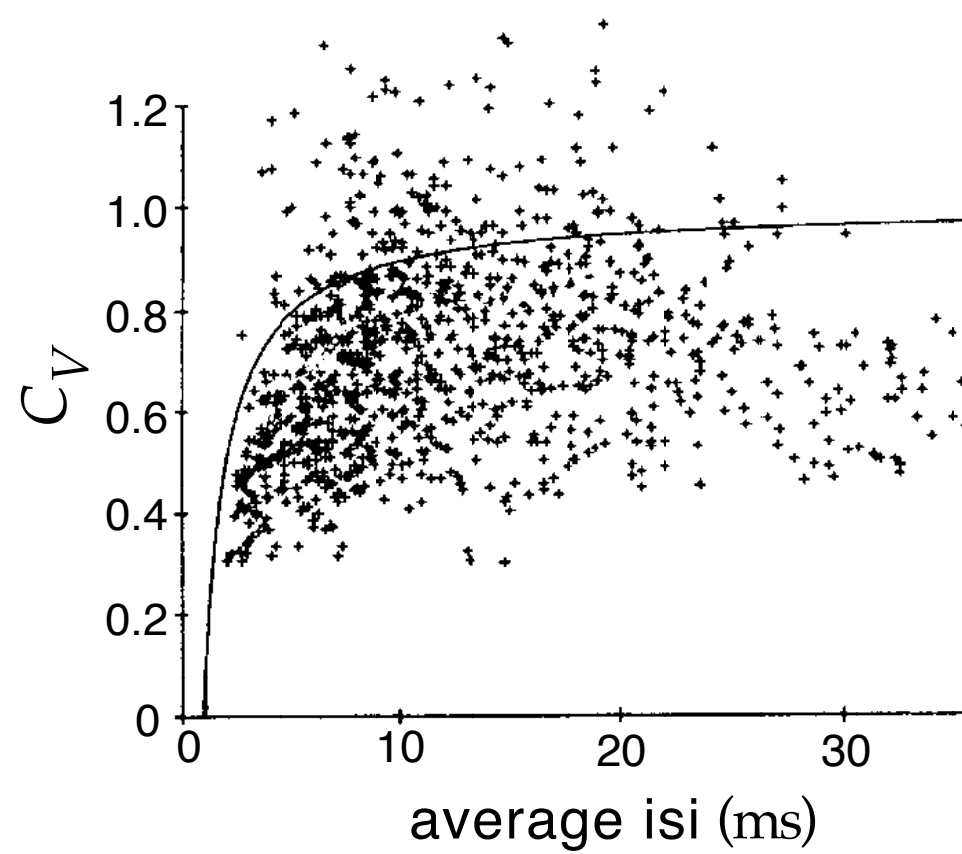


Figure 1.15: (A) Interspike interval distribution from an MT neuron responding to a moving random dot image. The probability of interspike intervals falling into the different bins, expressed as a percentage, is plotted against interspike interval. B) Interspike interval histogram generated from a Poisson model with a stochastic refractory period. (Adapted from Bair et al., 1994.)

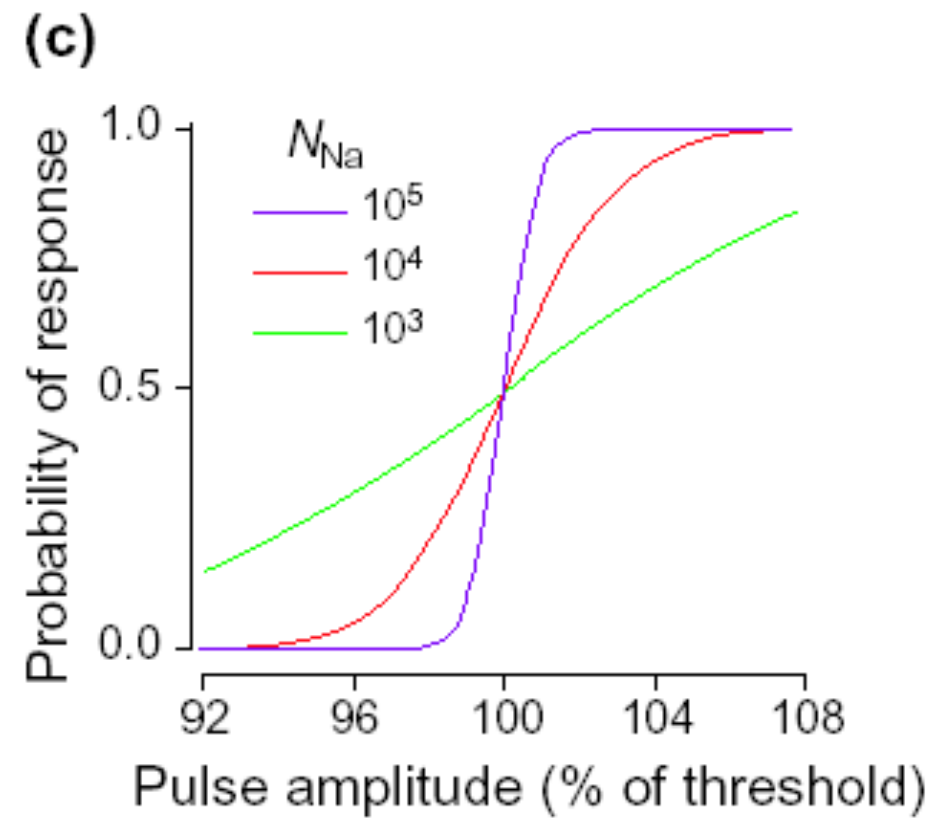
The Poisson spike train



Where does the variability come from?

- Channel noise
- Presynaptic sources are noisy

Impact of channel noise on spike generation threshold



Neurons *in vitro* respond reliably to fluctuating stimulus

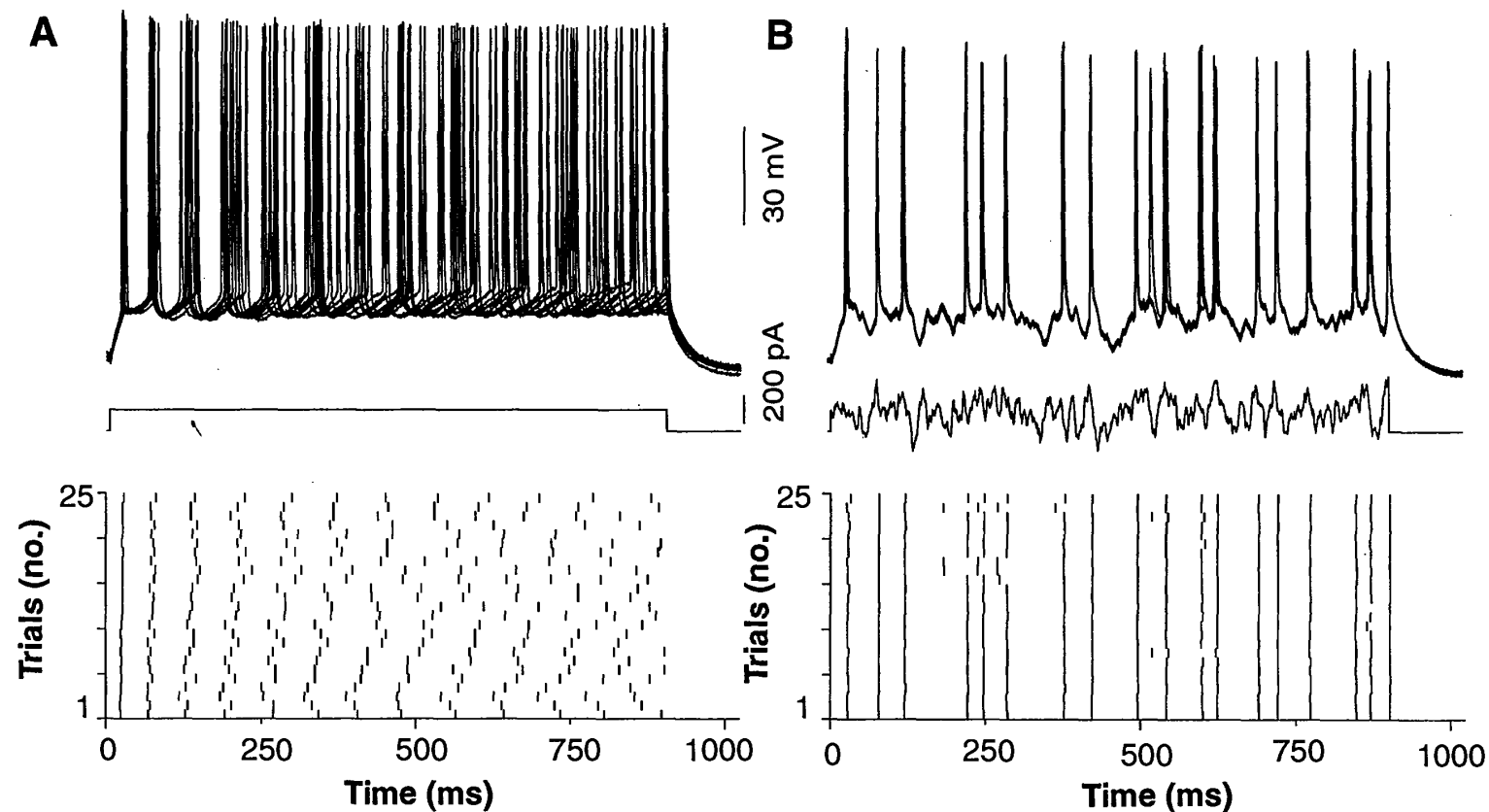


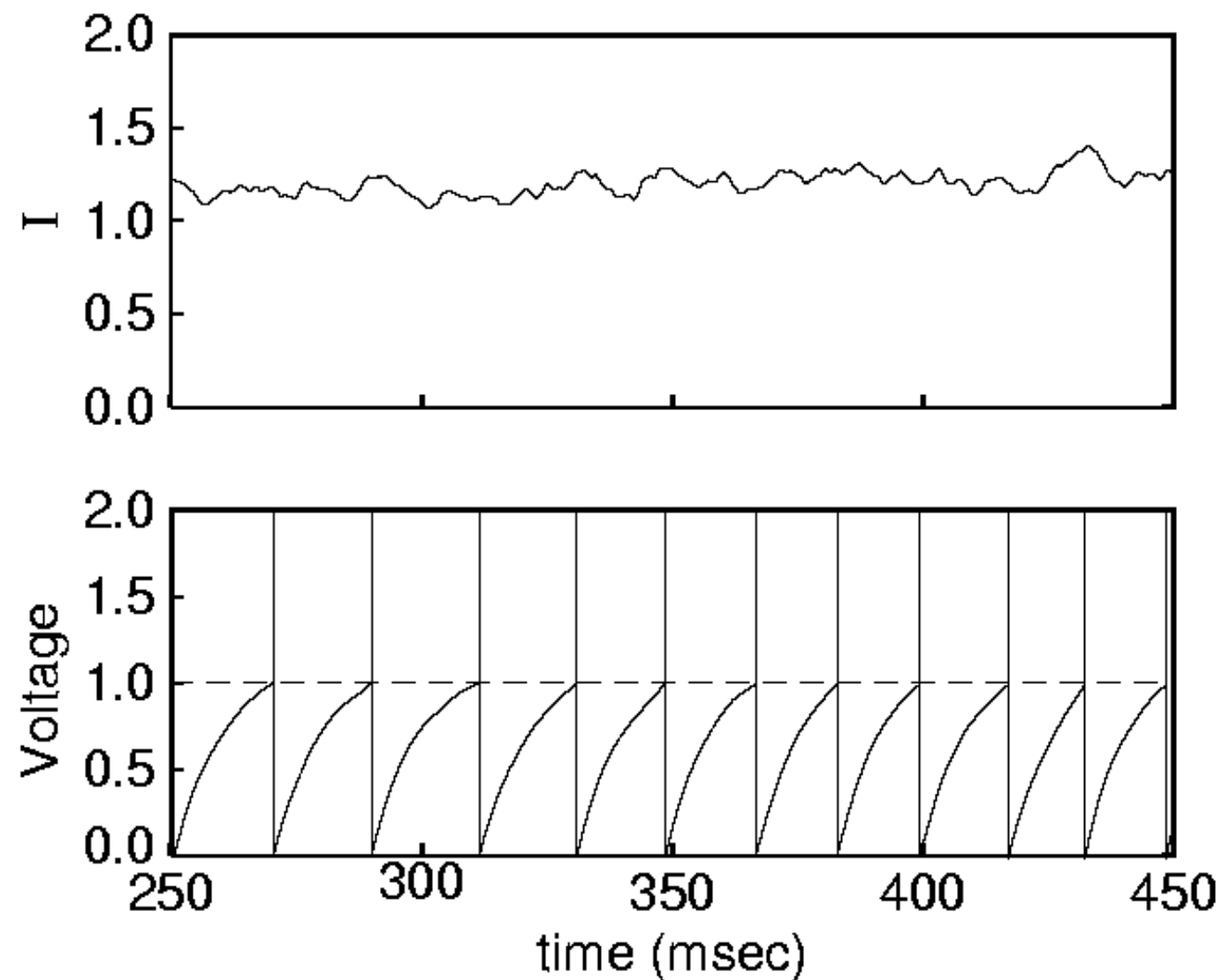
Fig. 1. Reliability of firing patterns of cortical neurons evoked by constant and fluctuating current. **(A)** In this example, a superthreshold dc current pulse (150 pA, 900 ms; middle) evoked trains of action potentials (approximately 14 Hz) in a regular-firing layer-5 neuron. Responses are shown superimposed (first 10 trials, top) and as a raster plot of spike times over spike times (25 consecutive trials, bottom). **(B)** The same cell as in (A) was again stimulated repeatedly, but this time with a fluctuating stimulus [Gaussian white noise, $\mu_s = 150$ pA, $\sigma_s = 100$ pA, $\tau_s = 3$ ms; see (14)].

Where does the variability come from?

- Channel noise
- Presynaptic sources are noisy

Simulation

Integrate and Fire Neuron with $K=1000$
uncorrelated Poisson synaptic inputs



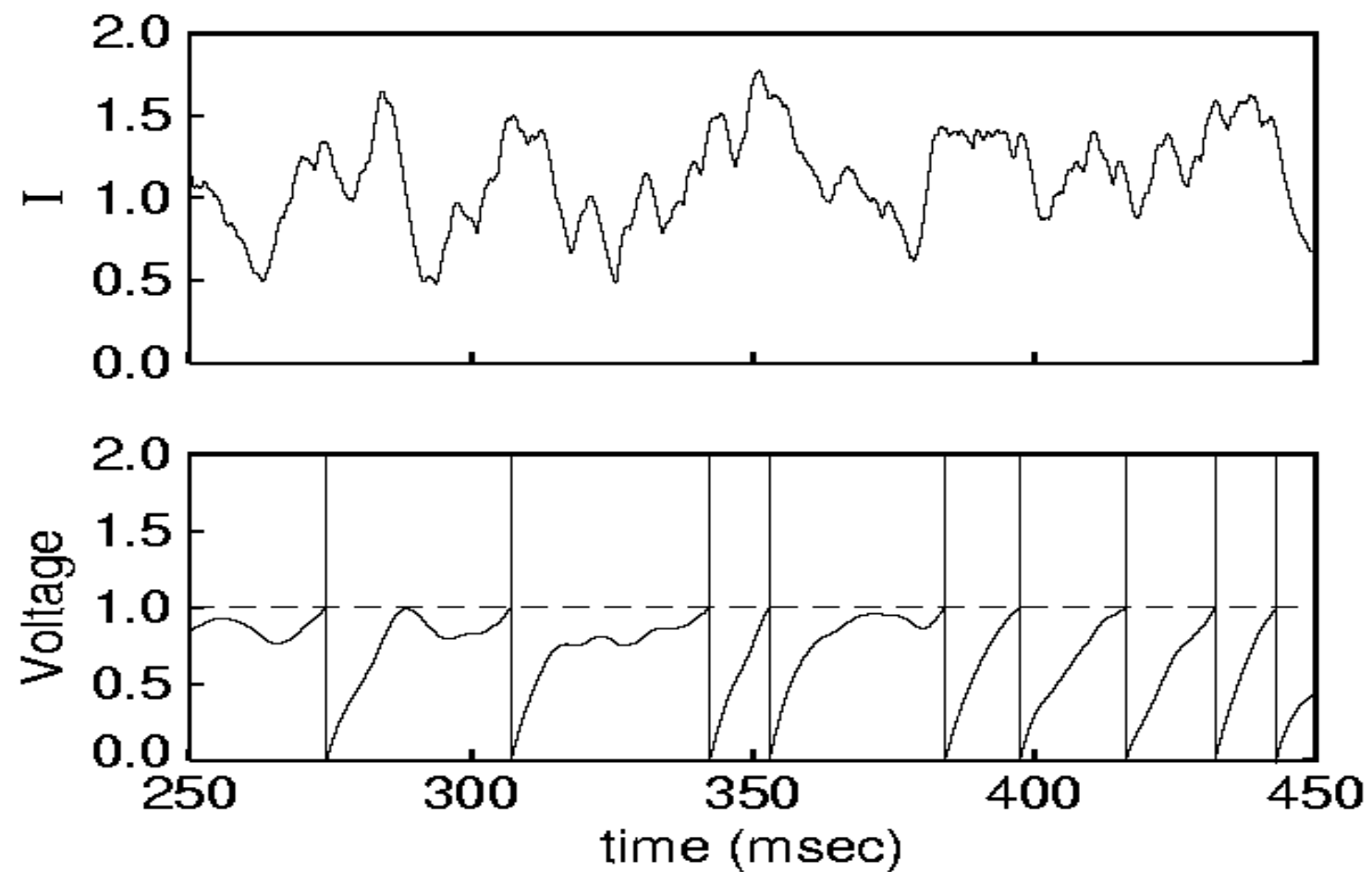
Two possible solutions

- Correlated Inputs
- Balanced Excitation and Inhibition to push the membrane potential near threshold

Simulation

Large Fluctuations due to Correlated Inputs

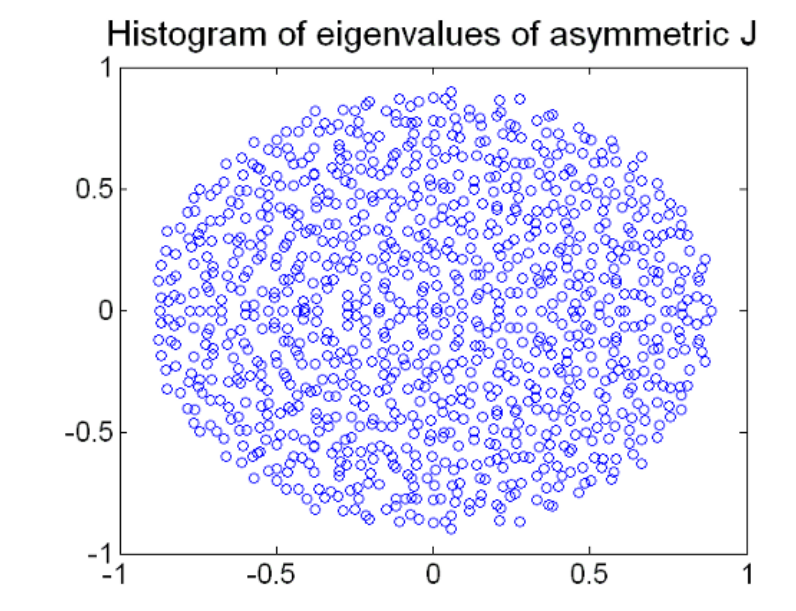
Integrate and Fire Neuron with $K=1000$
Poisson inputs with $c=0.1$ correlations



Interim: dynamics of random recurrent network

$$\tau \frac{dr_i}{dt} = -r_i + \tanh\left(\sum_{j=1}^N J_{ij} r_j\right)$$

$$\langle J_{ij} \rangle = 0, \quad \langle J_{ij}^2 \rangle = \frac{\tilde{J}^2}{N}$$



$$|\lambda|_{\max} = \tilde{J}$$

Dynamics of random recurrent network

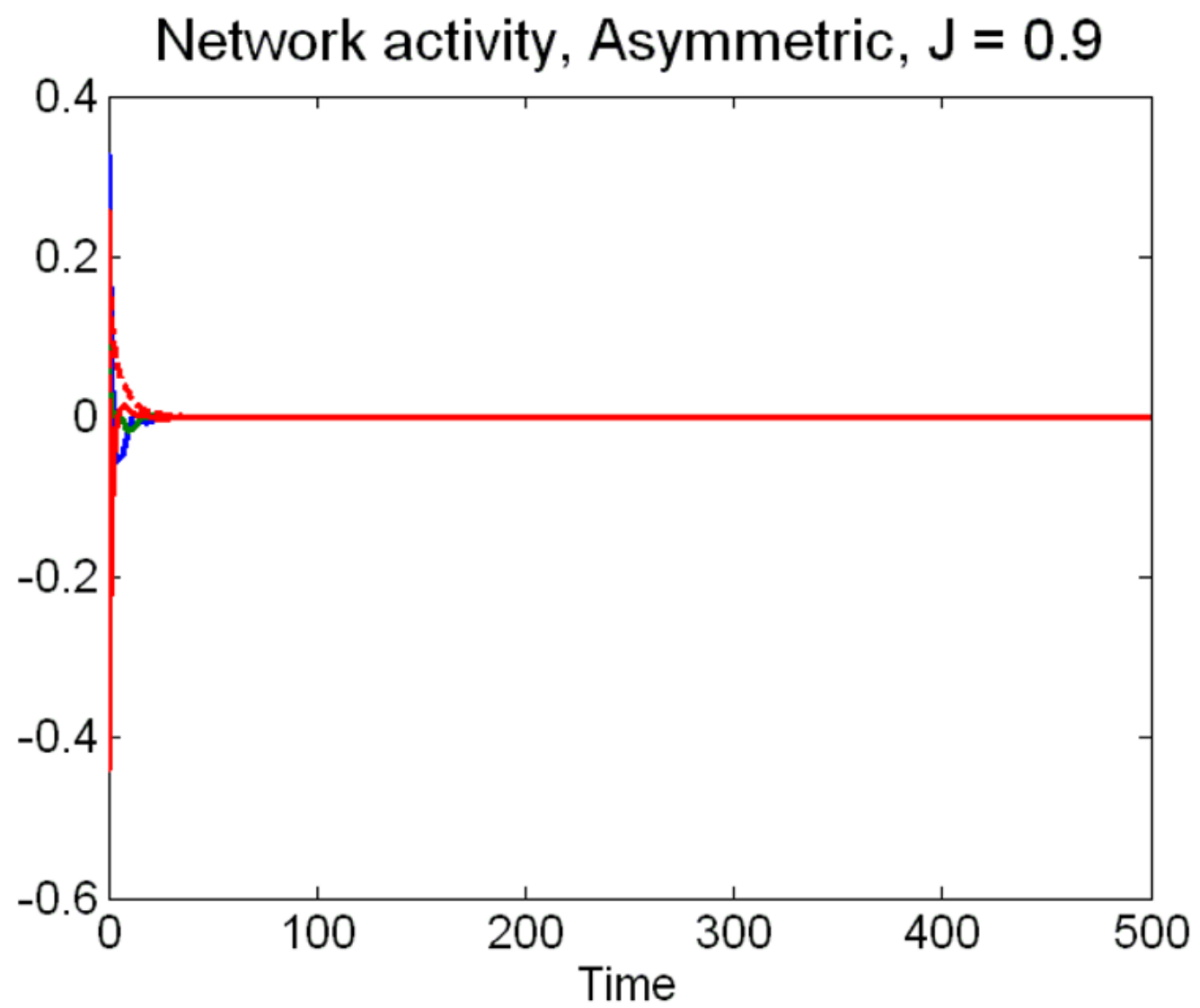
- Zero fixed point

Always exist, stable only $\tilde{J} < 1$

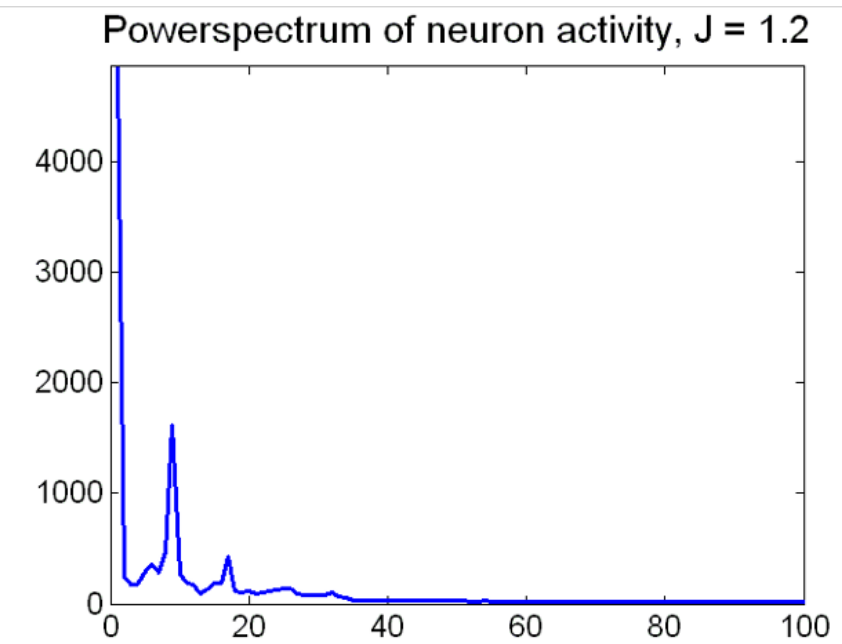
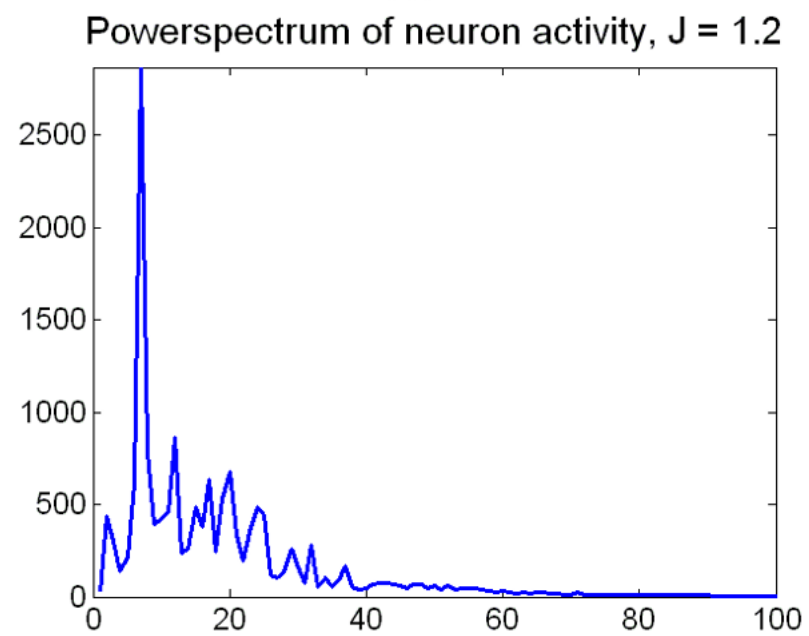
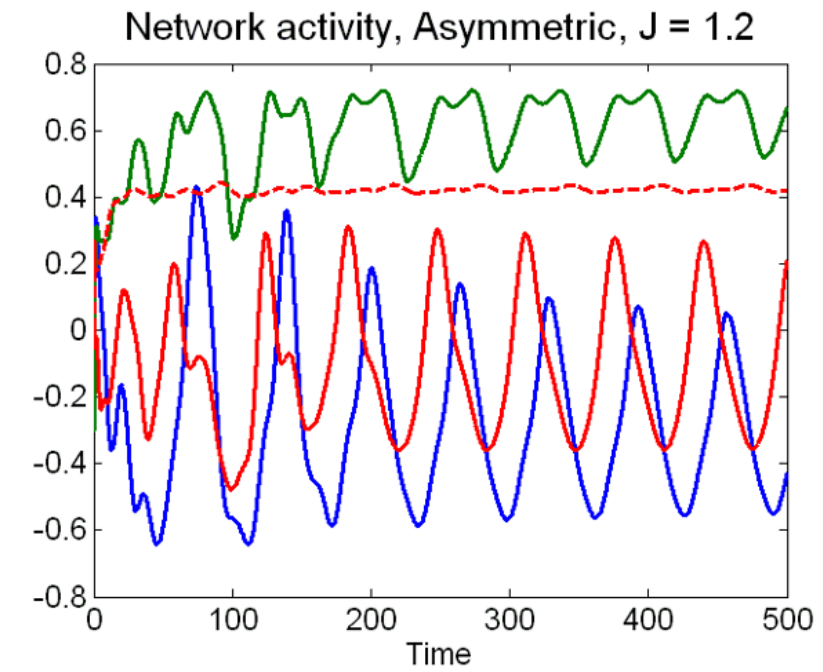
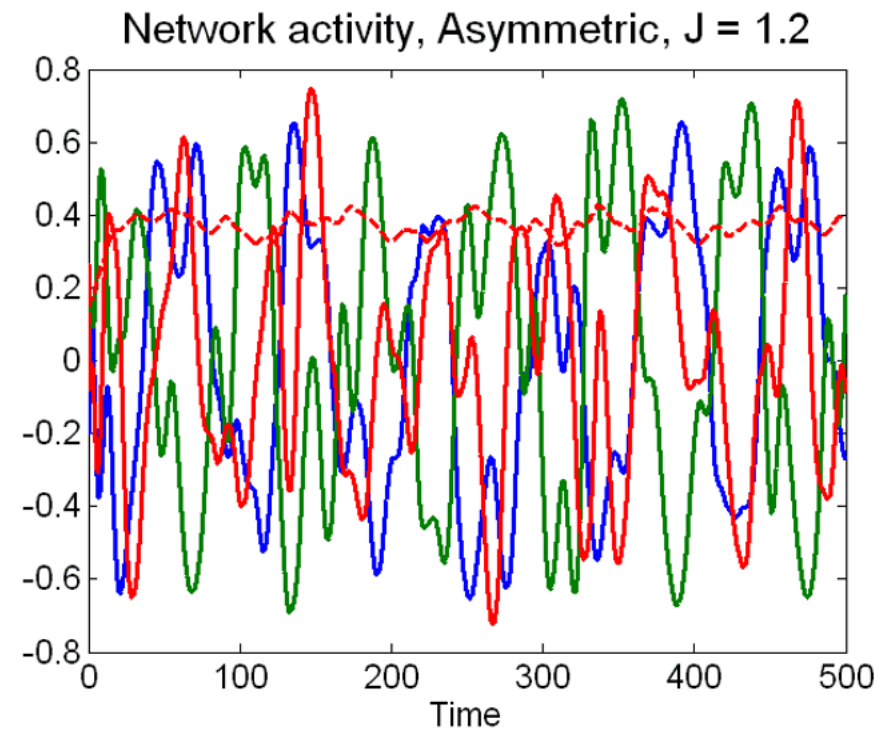
- Nontrivial attractor $\tilde{J} > 1$

For large N , chaotic attractor

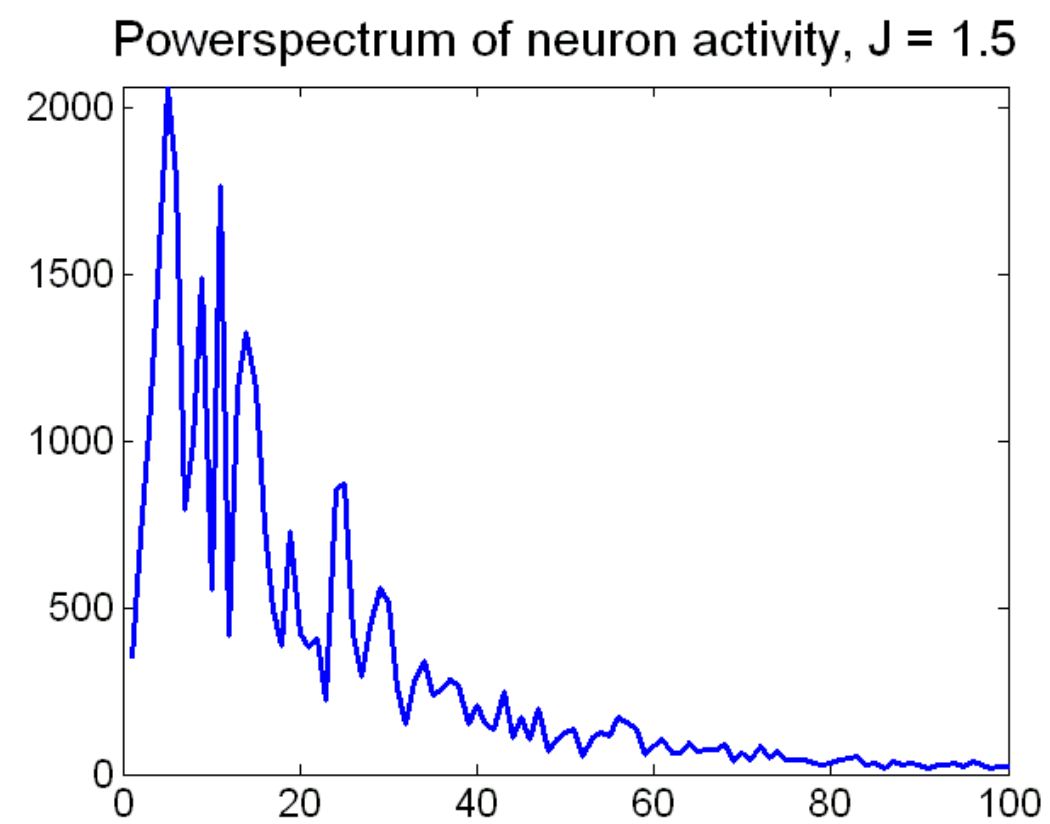
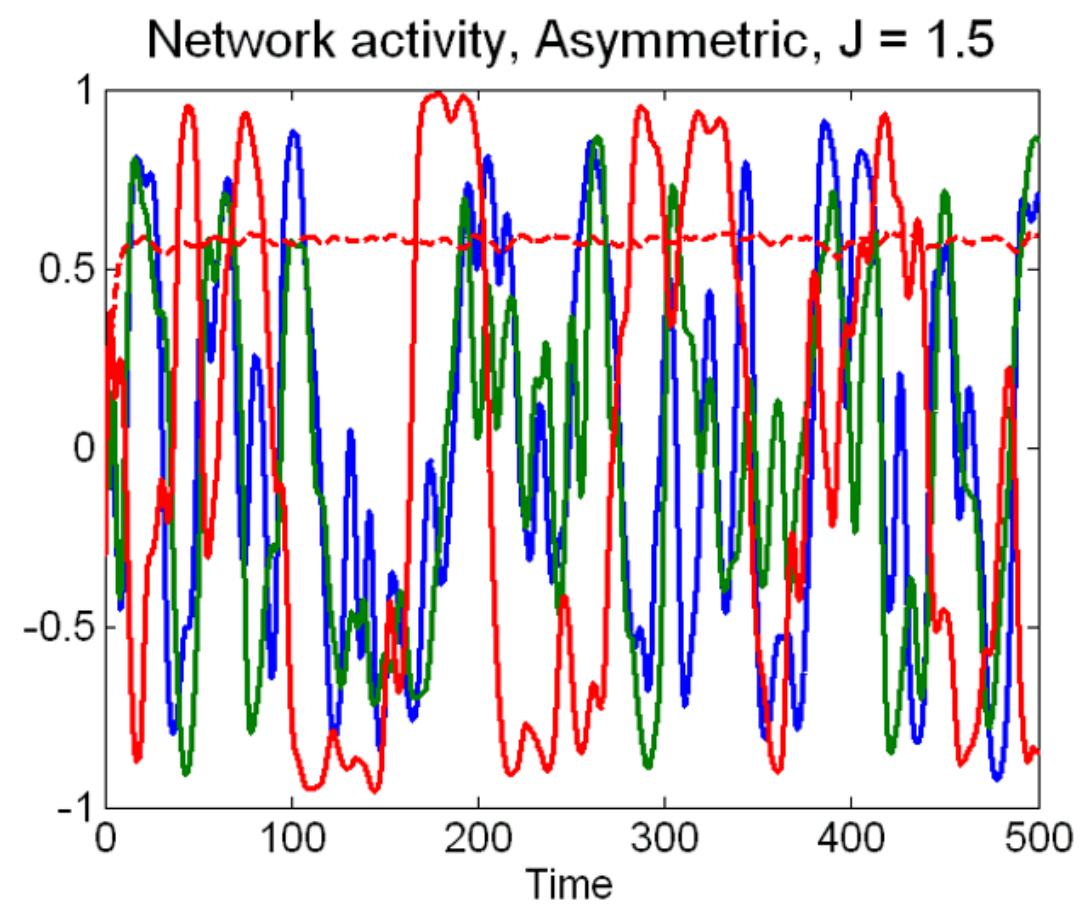
$$\tilde{J} < 1$$



$$\tilde{J} > 1$$

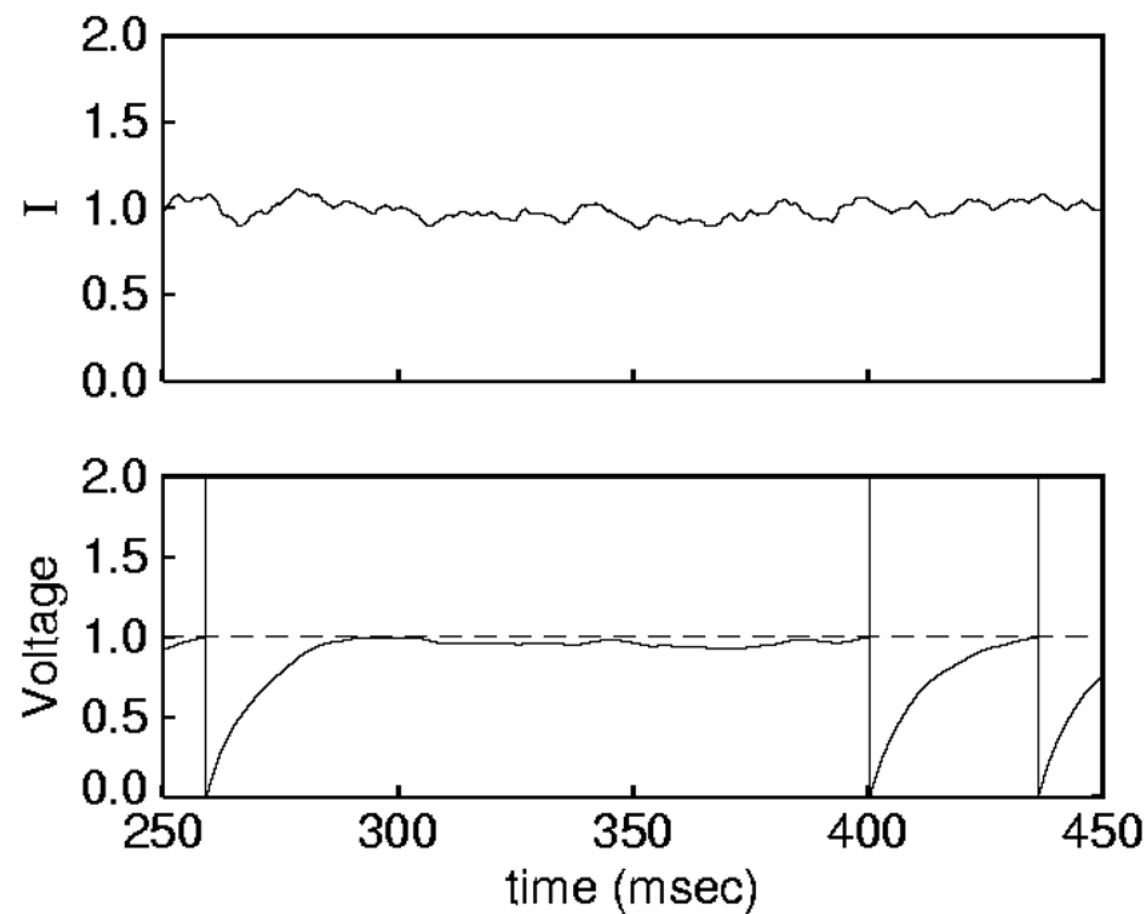


$$\tilde{J} > 1$$



Dale's rule and E-I balanced network

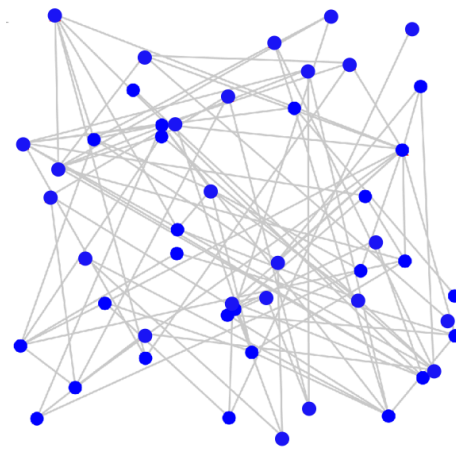
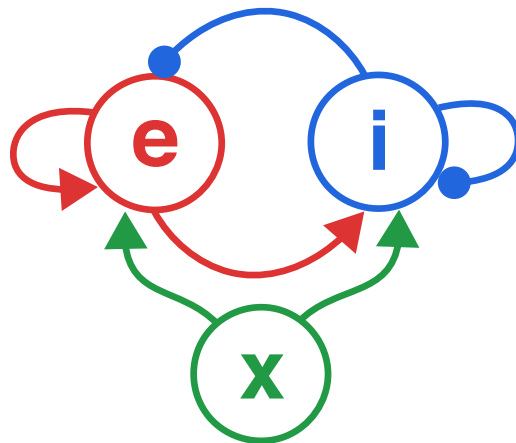
- Binary neurons, a toy model (whiteboard)



Dale's rule and E-I balanced network

- Firing rate model

Local cortical circuits



Randomly connected to K neurons

$$W_{ij}^{ee} = 0 \text{ or } \frac{w_{ee}}{\sqrt{K}}$$

(1) Individual connections are strong
(vs. weak connections $\frac{w_{ee}}{K}$)

(2) K is large

$$\tau \frac{dI_i}{dt} = -I_i + \sum_j W_{ij}^{ee} y_j^e(t) - \sum_j W_{ij}^{ei} y_j^i(t) + \sum_j W_{ij}^x y_j^x(t)$$

$$\tau \frac{dI_i}{dt} = -I_i + \sum_j W_{ij}^{ee} y_j^e(t) - \sum_j W_{ij}^{ei} y_j^i(t) + \sum_j W_{ij}^{ex} y_j^x(t)$$

average over time:

$$\left\langle \sum_j^{N_e} W_{ij}^{ee} y_j^e(t) \right\rangle_t = \sum_j^{N_e} W_{ij}^{ee} r_j^e$$

average over different neurons j :

$$\left\langle \sum_j^{N_e} W_{ij}^{ee} r_j^e \right\rangle_j \approx \sum_j^{N_e} \langle W_{ij}^{ee} \rangle \langle r_j^e \rangle$$

$$\langle W_{ij}^{ee} \rangle = \frac{K}{N_e} \frac{w_{ee}}{\sqrt{K}} \qquad \langle r_j^e \rangle = r^e$$

$$\left\langle \sum_j^{N_e} W_{ij}^{ee} r_j^e \right\rangle \approx \sqrt{K} w_{ee} r^e$$

average input to an excitatory neuron:

$$\left\langle \sum_j W_{ij}^{ee} y_j^e(t) - \sum_j W_{ij}^{ei} y_j^i(t) + \sum_j W_{ij}^{ex} y_j^x(t) \right\rangle_{t,j} \approx \sqrt{K}(w_{ee}r^e - w_{ei}r^i + w_{ex}r^x)$$

average input to an inhibitory neuron:

$$\sqrt{K}(-w_{ii}r^i + w_{ie}r^e + w_{ix}r^x)$$

Since the input should be $O(1)$, we must have the E-I balanced condition

$$\mathbf{E} : w_{ee}r^e - w_{ei}r^i + w_{ex}r^x = 0$$

$$\mathbf{I} : w_{ie}r^e - w_{ii}r^i + w_{ix}r^x = 0$$

$$r^e \geq 0, r^i \geq 0, r^x \geq 0$$

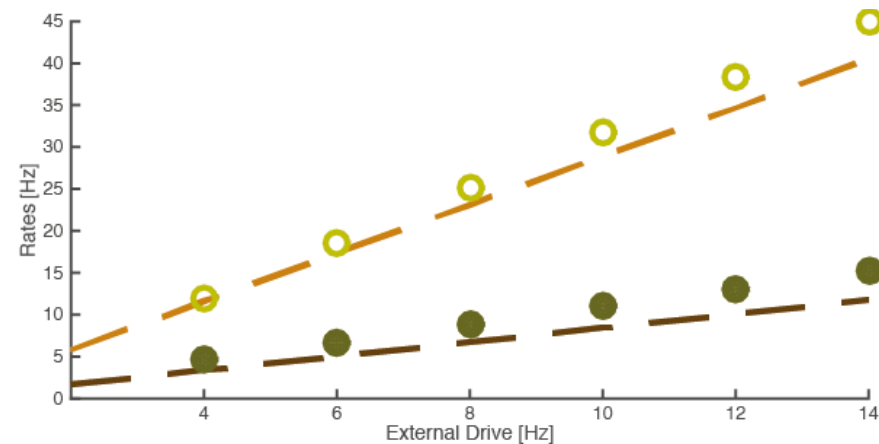
Tight E-I balance condition

$$E \quad w_{ee}r^e - w_{ei}r^i + w_{ex}r^x = 0$$

$$I \quad w_{ie}r^e - w_{ii}r^i + w_{ix}r^x = 0$$

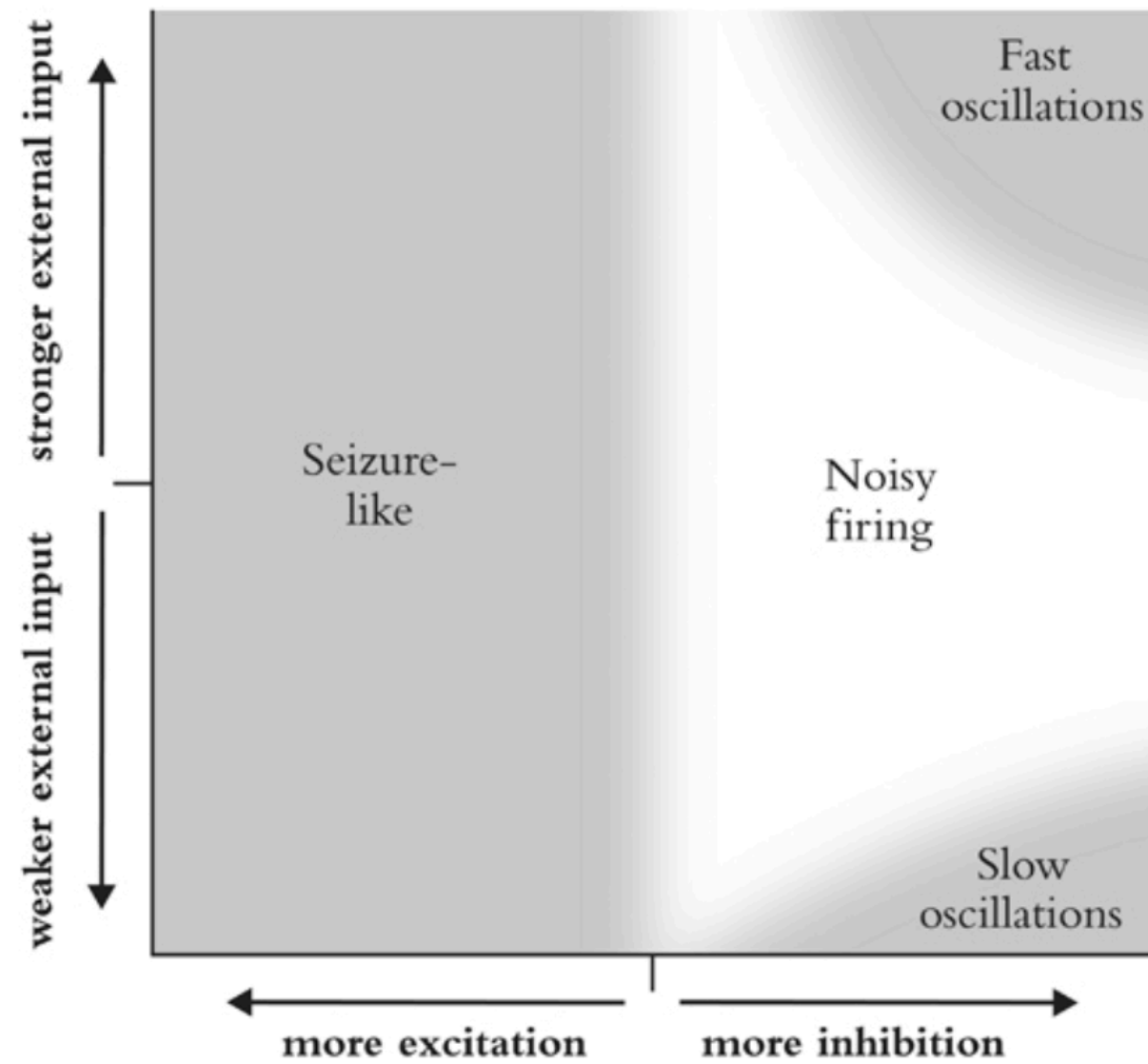
- Solve for population firing rates $r = -W^{-1}w_x r_x$
- Predict linear population response to input!

E, filled and **green**
I, open and **yellow**

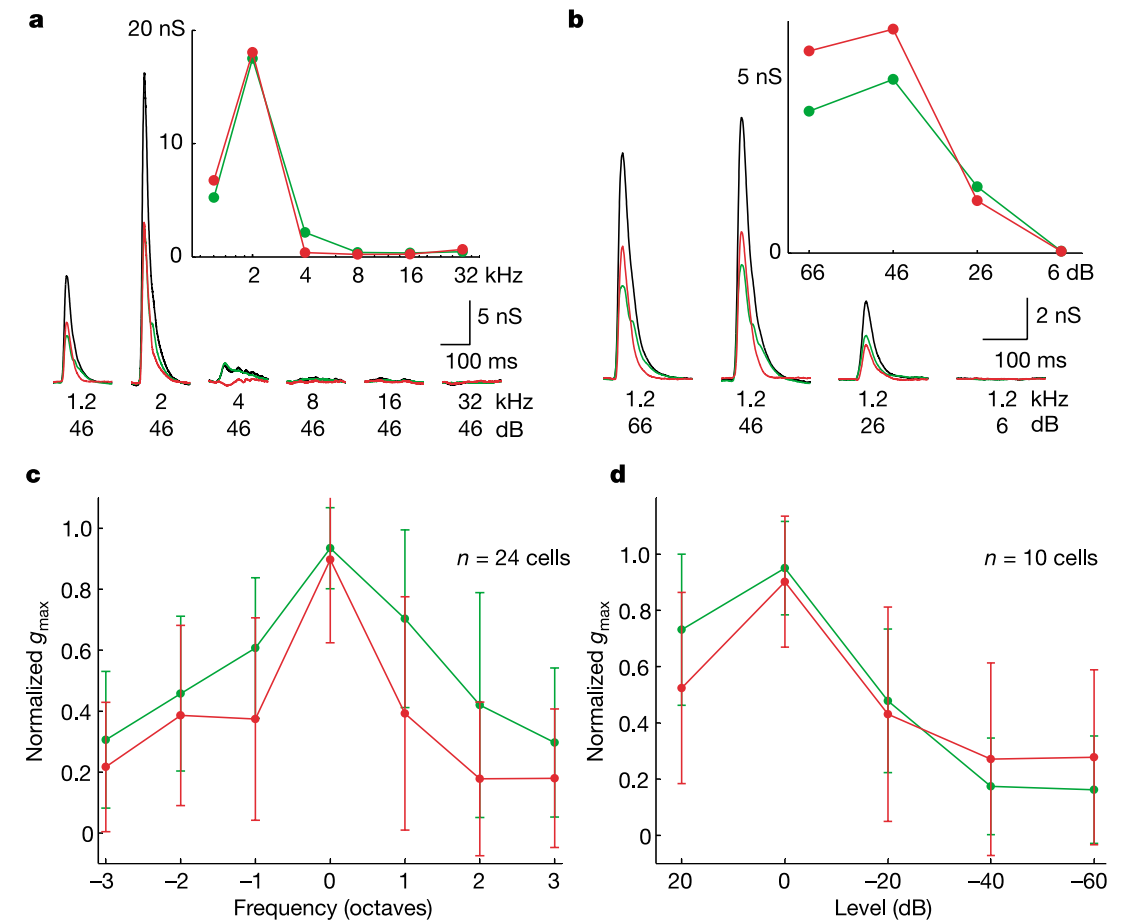
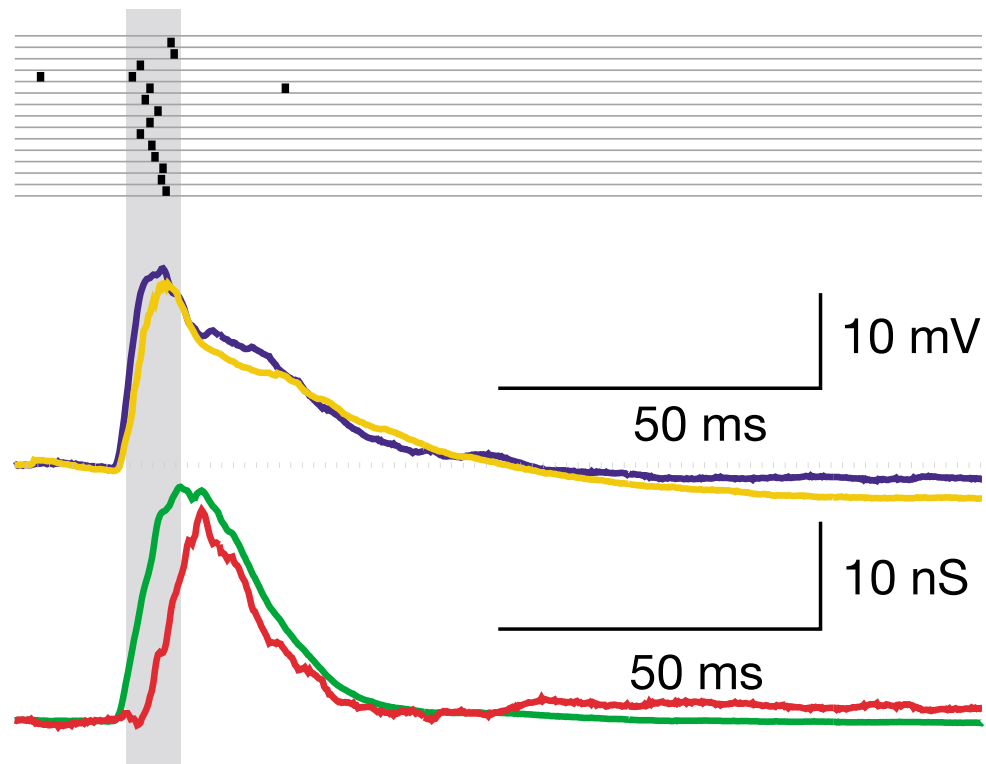


Vreeswijk and Sompolinsky 1996

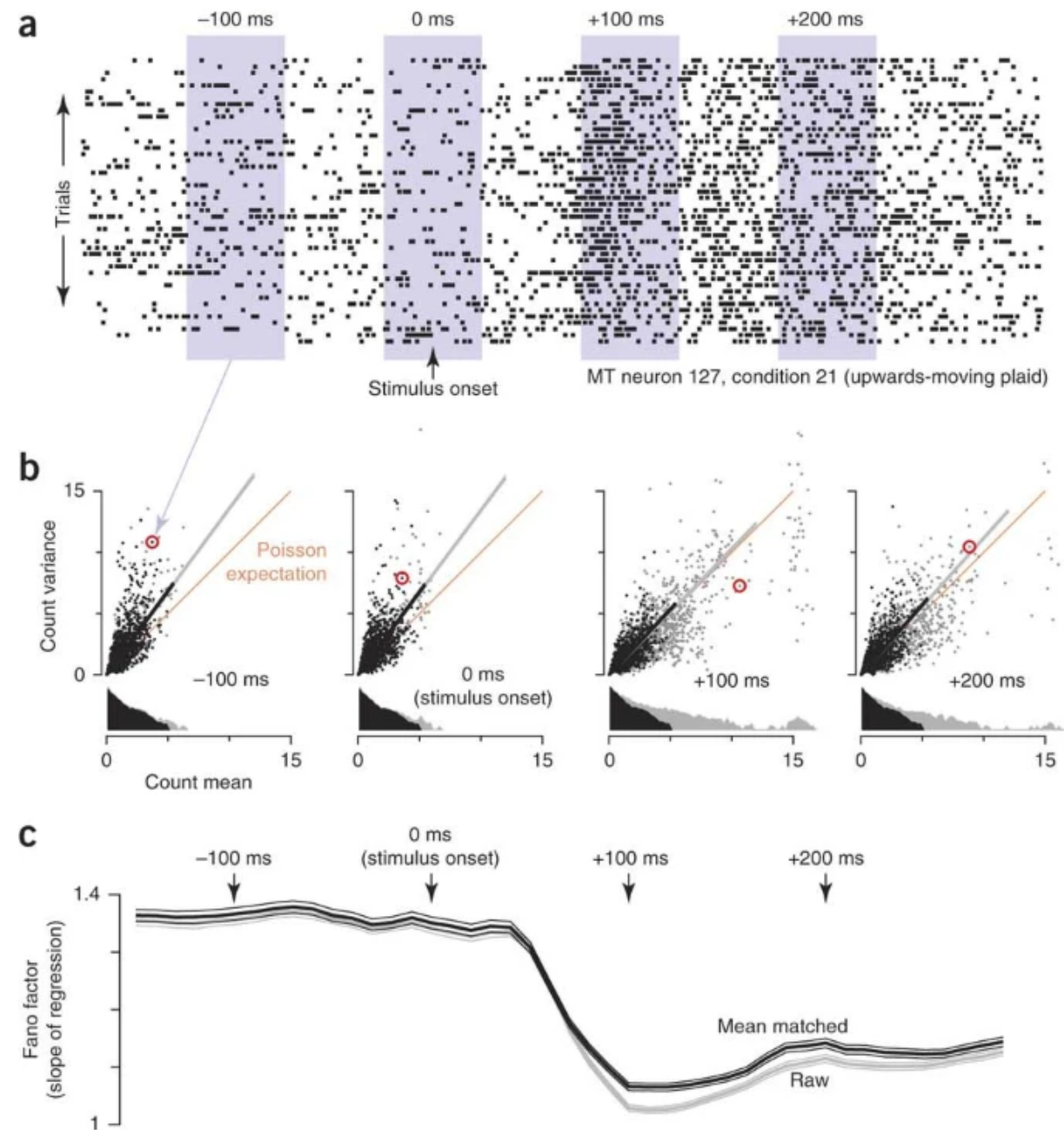
The phase diagram



Balanced inhibition in auditory cortical neurons

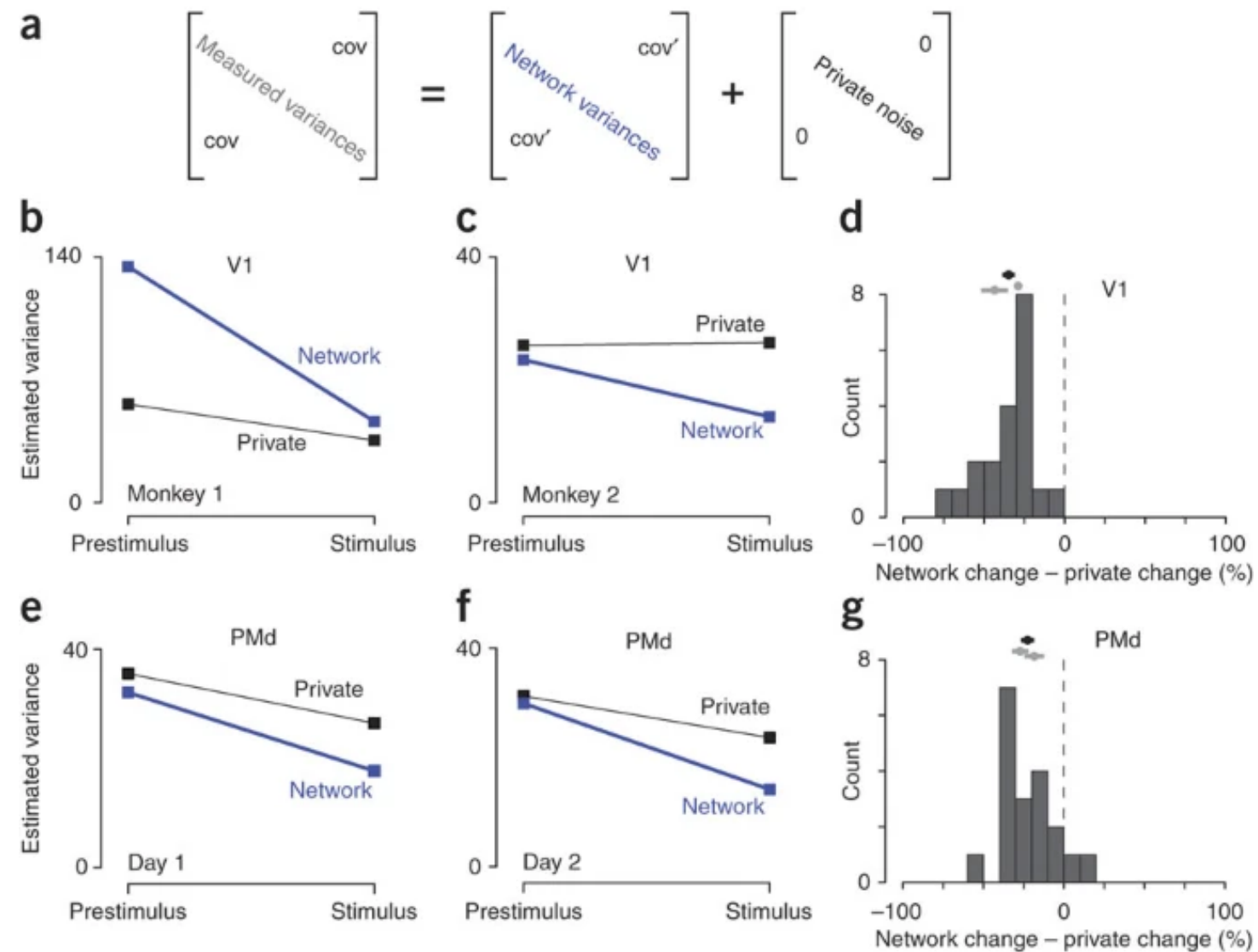


Quenched neural variability at stimulus onset



Churchland *et al* 2010

Reduced correlated neural variability at stimulus onset



Neuron



Perspective

What is the dynamical regime of cerebral cortex?

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²Center for Theoretical Neuroscience, Swartz Program in Theoretical Neuroscience, Kavli Institute for Brain Science, and Department of Neuroscience, College of Physicians and Surgeons and Morton B. Zuckerman Mind Brain Behavior Institute, Columbia University, New York, NY, USA

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